

- ▶ You can combine multiple hard drives, whether internal, or external, connected via USB, SATA, or SAS. When you create a storage pool, all the storage in that pool of drives is combined.
- ▶ Within this storage pool you can create storage spaces. A storage space is like a partition on a hard drive, it has a certain capacity, and can store files up to that capacity.
- ▶ A storage space can be of four resiliency types:
 - **Two-way mirror:** this keeps each piece of data on at least two different drives. For this you need at least two drives in the pool.
 - **Three-way mirror:** this quite expectedly keeps all data on three drives. For this you need at least three drives in the pool.
 - **Parity:** This is more complex; it stores a copy of the data, and enough information about the data to reconstruct it if one of your drives fails. For this too you need at least three drives in the pool.
 - **Simple:** This is the standard way of doing things, just one copy of the data. You only need a single drive in a pool for this.
- ▶ A storage space is not limited by the size of the drives in the pool. This means that even if you only have three 1TB drives in a pool, you can still make a 10TB storage space in that pool. You can't really use that space, but as you add more drives to the pool the space will become available.
- ▶ A single storage pool can have multiple spaces of different types. So if you group 4x1TB drives in a pool, it can have a two-way mirrored pool of 5TB, another three-way mirrored space of 1TB, a 10TB parity space and a 5TB simple space.
- ▶ Since data in a pool is stored in multiple places (except for simple storage spaces) if one of your drives fails you can disconnect it, add a new drive, and the data will be rebuilt on that new drive. So even if a drive doesn't fail you can remove your old 250GB drive and plug in a new 3TB drive to increase the size of your pool.
- ▶ Even if you just create a simple storage space, at least you can combine the storage space of multiple drives in one place. So rather than have "Video", "Video 2" and so on, you have one storage space that you keep expanding with more drives.

To actually use this feature simply visit Storage Spaces in your Control Panel, you can also find Storage Spaces by pressing [Win]+[W] to launch settings search, typing "storage space" and pressing Enter. You can then connect the devices you would like to create a pool with—or if they are internal drives, they're already connected.